

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO1

Paper: Sample Multiple Choice Questions - ANSWER KEY

Maximum Marks: 40 x 1 = 40

Unit: 1

Unit Title: Foundations of Unsupervised Learning

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the measure that gives the straight-line gap between two records.

- A) Euclidean distance <-- correct
- B) Manhattan distance
- C) Cosine similarity
- D) Cosine distance

Answer: A. Euclidean distance is the square root of the summed squared differences, which is the ordinary straight-line distance between two points.

Q2. State the range of values cosine similarity can take when the vectors may hold negative entries.

- A) From 0 to 1 only
- B) From -1 to +1 <-- correct
- C) From 0 to infinity
- D) From -infinity to +infinity

Answer: B. Cosine similarity is the cosine of the angle between the vectors, so it lies between -1 and +1. It is confined to 0 to 1 only when every entry is non-negative.

Q3. Define cosine distance in terms of cosine similarity.

- A) The reciprocal of the cosine similarity
- B) The square of the cosine similarity
- C) One minus the cosine similarity <-- correct
- D) The negative of the cosine similarity

Answer: C. Cosine distance is defined as 1 - cosine similarity, which turns a measure that is large for similar records into one that is small for them.

Q4. Name the stage of an analysis in which the data is summarised and plotted before any model is built.

- A) Model deployment
- B) Hyperparameter tuning
- C) Cross validation
- D) Exploratory data analysis <-- correct

Answer: D. Exploratory data analysis is the inspection stage: distributions, ranges, missing values and relationships are examined before any method is chosen.

Q5. Identify what a clustering algorithm produces for each record of the input table.

- A) A predicted numeric value
- B) A cluster label <-- correct
- C) A probability of fraud
- D) A confidence interval

Answer: B. The output of a clustering is a group assignment for every record. The labels themselves are arbitrary names for the groups.

Q6. Name the unsupervised task that finds tightly connected groups of users inside a social network.

- A) Sentiment analysis
- B) Link prediction
- C) Spam classification

D) Community detection <-- correct

Answer: D. Community detection groups the nodes of a network by how densely they are connected, which is pattern discovery in social data.

Q7. Define semi-supervised learning.

- A) Learning from a few labelled records and many unlabelled ones <-- correct
- B) Learning from labelled records only
- C) Learning from a reward given after each action
- D) Learning from records that have two target columns

Answer: A. Semi-supervised learning combines a small labelled sample with a large unlabelled one, and is used when labelling every record would be too costly.

Q8. State the value a Euclidean distance takes when the two points are identical.

- A) One
- B) Infinity
- C) Zero <-- correct
- D) It is undefined

Answer: C. Every coordinate difference is zero, so the sum of squares is zero and so is its square root. A distance of zero means the two records coincide.

Q9. Name the three measures listed on the syllabus for comparing a pair of records.

- A) Inertia, silhouette and the Davies-Bouldin Index
- B) Support, confidence and lift
- C) Euclidean distance, Manhattan distance and cosine similarity <-- correct
- D) Accuracy, precision and recall

Answer: C. These three are the similarity and distance measures of Unit 1. The second set belongs to cluster validation and the third to supervised learning.

Q10. Name the quantity that fixes the number of dimensions of the space in which the records lie.

- A) The number of features <-- correct
- B) The number of records
- C) The number of clusters
- D) The number of iterations

Answer: A. Each feature is one axis, so a table of eight features places every record in an eight-dimensional space.

Q11. State what the target column of a supervised dataset holds.

- A) The row number of each record
- B) The distance to the nearest record
- C) The cluster each record belongs to
- D) The correct answer for each record <-- correct

Answer: D. The target, or label, is the known correct output. Its absence is exactly what makes a dataset unsupervised.

Q12. Identify the operation Euclidean distance applies to the sum of the squared differences.

- A) The absolute value
- B) The square root <-- correct
- C) The natural logarithm
- D) The reciprocal

Answer: B. Euclidean distance is the square root of the summed squared differences, which returns the answer to the same units as the features.

Q13. Name the type of feature that records a category rather than a number.

- A) A continuous feature
- B) A derived feature
- C) A target feature
- D) A categorical feature <-- correct

Answer: D. A categorical feature takes one of a fixed set of values, such as a city or a payment method, and must be encoded before a distance can be computed.

Q14. Recall what knowledge discovery in an unlabelled dataset sets out to produce.

- A) A forecast of next month's revenue
- B) A label for every record in the table
- C) Patterns that were not known in advance <-- correct
- D) A ranking of the features by importance

Answer: C. Knowledge discovery is the search for structure nobody specified beforehand, which is why the task cannot be posed as predicting a known target.

Q15. State what must be chosen before any distance-based unsupervised method can be run.

- A) A target column to predict
- B) A measure of distance between records <-- correct
- C) A training and a test partition
- D) A confidence level for the result

Answer: B. The algorithm sees the data only through the distances, so the measure must be settled first. Everything the method reports follows from that choice.

Q16. Identify the term used for a single row of an unlabelled dataset.

- A) A record <-- correct
- B) A cluster
- C) A centroid
- D) A feature

Answer: A. A record, also called an instance or observation, is one row. A feature is one column, and a cluster is a group of records.

Q17. Explain why cosine similarity is unsuitable when the total amount a customer spends is the quantity of interest.

- A) It cannot be computed on spending data
- B) It ignores the size of the vectors entirely <-- correct
- C) It always returns a value close to zero
- D) It requires the number of groups in advance

Answer: B. Cosine reads only the angle, so a customer spending 500 and one spending 50,000 in the same proportions look identical to it. Any question about how much is spent needs a magnitude-sensitive measure.

Q18. Distinguish a clustering task from a classification task by what each one is given.

- A) Clustering is given no labels, classification is given labels <-- correct
- B) Clustering is given labels, classification is given none
- C) Both of the two are given a full set of labels
- D) Both of the two are given a single labelled record

Answer: A. Classification learns to reproduce known labels; clustering has none and must find the groups from the features alone.

Q19. Predict the effect on a Manhattan distance of adding one extra feature whose value is identical in both records.

- A) The distance is doubled
- B) The distance falls to zero
- C) The distance can no longer be computed
- D) The distance is left unchanged <-- correct

Answer: D. The new coordinate contributes an absolute difference of zero to the sum, so the total is unaffected. Only features that differ can move a distance.

Q20. Interpret a cosine similarity of exactly zero between two purchase vectors holding non-negative counts.

- A) The two customers bought identical baskets
- B) The two customers spent the same amount
- C) The two customers bought no item in common <-- correct
- D) One of the two customers bought nothing

Answer: C. With non-negative entries the similarity is zero only when the vectors are orthogonal, which happens when no product has a non-zero count in both.

Q21. Predict what happens to a Euclidean distance when forty uninformative random features are added to a table.

- A) The genuine differences are drowned out <-- correct
- B) The genuine differences are sharpened
- C) The distance becomes exactly zero
- D) The distance stops depending on the data

Answer: A. Every added feature contributes noise to the sum, so the contrast between a near pair and a far pair shrinks and the grouping degrades.

Q22. Recognize the unsupervised task described: a telecom operator groups its cell towers by their hourly traffic profile so that capacity can be planned tower by tower.

- A) Supervised regression on traffic
- B) Classification of tower faults
- C) Clustering for segmentation <-- correct
- D) Reinforcement learning of a policy

Answer: C. No group labels exist beforehand; the towers are divided by similarity of their traffic profiles, which is a clustering task.

Q23. Summarize why labels are frequently unavailable in a real project.

- A) Labels are forbidden by data protection law
- B) Labelling is slow, costly and often needs an expert <-- correct
- C) Labels cannot be stored in a numeric table
- D) Labels make a model less accurate overall

Answer: B. Someone has to decide and record the correct answer for each record. On millions of rows, or where the judgement needs a specialist, this is impractical.

Q24. Explain why two customers with identical spending patterns but very different totals sit far apart under Euclidean distance.

- A) Euclidean distance only compares the largest feature
- B) Euclidean distance cannot handle spending data
- C) Euclidean distance ignores the smaller of the two
- D) Euclidean distance grows with the size of the values <-- correct

Answer: D. The gap on every axis is proportional to the difference in totals, so a large difference in scale produces a large distance even when the shape of the spending matches.

Q25. Distinguish a Euclidean distance of zero from a cosine similarity of zero.

- A) Zero distance means unrelated, zero cosine means identical
- B) Both of the two mean that the records are identical
- C) Zero distance means identical, zero cosine means unrelated <-- correct
- D) Both of the two mean that the records are unrelated

Answer: C. A distance of zero says the two records coincide. A cosine of zero says the two vectors are at right angles, which means they share no direction at all.

Q26. Interpret a cosine similarity of -1 between two feature vectors.

- A) They are identical to one another
- B) They share no feature in common
- C) One of the two is the zero vector
- D) They point in exactly opposite directions <-- correct

Answer: D. A cosine of -1 is an angle of 180 degrees. It can only arise when negative entries are present, as with standardised features.

Q27. Explain the difficulty a categorical feature introduces into a Euclidean distance.

- A) Every pair of different categories ends up equally far apart <-- correct
- B) The distance can no longer be computed at all
- C) The distance becomes negative for some of the pairs
- D) The two categories are silently merged into one

Answer: A. Once a category is turned into indicator columns, any two different categories differ in the same way, so a near-miss and a completely unrelated category count as equally dissimilar.

Q28. Predict the effect on a clustering of removing the one feature that carried the real structure.

- A) The groups become sharper than before
- B) The groups it revealed can no longer be found <-- correct
- C) The algorithm refuses to run at all
- D) The number of groups doubles automatically

Answer: B. The method sees only the columns it is given. Structure that lived in a discarded feature is simply invisible to it.

Q29. Describe what the formulation of an unsupervised problem has to settle.

- A) What accuracy the model must reach
- B) How many records are held back for testing
- C) Which label is attached to each record
- D) What is measured and what counts as similar <-- correct

Answer: D. With no target to predict, the two decisions that fix the result are the features recorded and the definition of similarity applied to them.

Q30. Recognize the situation in which semi-supervised learning is the natural choice.

- A) Every record carries a reliable label
- B) A few records are labelled and very many are not <-- correct
- C) No record can ever be labelled at all
- D) Every record carries two different labels

Answer: B. Semi-supervised methods let the few labels guide the structure found in the many unlabelled records, which is common when labelling is expensive.

Q31. Explain why the result of an unsupervised method has to be reviewed by someone who knows the domain.

- A) The algorithm cannot compute the distances reliably
- B) The software licence requires a human reviewer
- C) There is no ground truth to check the result against <-- correct
- D) The records must be relabelled before reporting

Answer: C. Nothing in the data says whether a grouping is right. Only somebody who understands the subject can judge whether the groups mean anything.

Q32. Distinguish exploratory data analysis from model building.

- A) Exploration describes the data, modelling acts on it <-- correct
- B) Exploration predicts values, modelling plots them
- C) Both of the two produce a fitted model
- D) Both of the two require a labelled target

Answer: A. Exploration is about understanding what is in the table. Modelling then uses that understanding to group, reduce or predict.

Q33. Explain why Manhattan distance is also called city-block distance.

- A) Movement is measured along the axes, not diagonally <-- correct
- B) It can only be used on map coordinates
- C) It was first devised for planning cities
- D) It measures the straight line between points

Answer: A. The sum of the axis-wise gaps is the distance a walker would cover on a rectangular street grid, where a diagonal shortcut is not available.

Q34. Explain why fraud detection is often treated as an unsupervised problem even when a few past frauds are on record.

- A) Past fraud records are legally unusable
- B) Fraud records cannot be stored numerically
- C) A labelled model would run far too slowly
- D) New fraud patterns rarely resemble the recorded ones <-- correct

Answer: D. A supervised model learns only the frauds it has seen. Treating the task as finding records unlike the normal bulk allows a previously unseen pattern to be flagged.

Q35. Predict what a clustering algorithm reports on a dataset that contains no genuine group structure.

- A) It reports that no clusters were found
- B) It still returns the number of groups it was asked for <-- correct
- C) It refuses to run and raises an error
- D) It returns every record as its own cluster

Answer: B. k-Means and its relatives always partition the data. A result is therefore never in itself evidence that structure exists, which is why validation matters.

Q36. Recognize the fault described: an analyst clusters a customer table in which the postcode is stored as a plain number, and obtains groups that follow numeric closeness in that column.

- A) The number of clusters was set too high
- B) The distance measure was chosen too late
- C) A categorical code has been treated as a numeric quantity <-- correct
- D) The table held too few records to cluster

Answer: C. Postcodes that are numerically near need not be geographically or behaviourally near. A code must be encoded as a category, or replaced by real coordinates.

Q37. Interpret a large Euclidean distance between two customers who purchased exactly the same set of products.

- A) They have no product in common at all
- B) One of the two bought nothing whatever
- C) They differ in quantity rather than in what they bought <-- correct
- D) The distance has been computed incorrectly

Answer: C. The same products with very different counts give large gaps on each axis. Cosine similarity would report these two as close, which is the point of the choice.

Q38. Summarize why the choice of distance can matter more than the choice of algorithm.

- A) The algorithm is chosen automatically by the software
- B) The algorithm sees the data only through the distances <-- correct
- C) Every algorithm produces exactly the same groups
- D) The distance decides how many records are kept

Answer: B. Once the distances are fixed, the different methods largely agree on what is close. Changing the measure changes the input to all of them at once.

Q39. Explain why a similarity measure and a distance measure move in opposite directions.

- A) A similarity is always negative in value
- B) A distance can never exceed the value one
- C) A similarity is computed on the labels only
- D) Records that are near are the records that are alike <-- correct

Answer: D. A small distance and a high similarity both say the same thing about a pair of records, which is why cosine distance is written as 1 minus cosine similarity.

Q40. Predict the effect on a market segmentation of including the customer's account number as a feature.

- A) It adds meaningless variation that can distort the groups <-- correct

- B) It sharpens the boundaries between the groups
- C) It guarantees that every group is the same size
- D) It has no effect of any kind on the result

Answer: A. An account number is an arbitrary identifier carrying no behavioural meaning, yet the distance treats it as a real quantity. Identifiers must be dropped before clustering.

----- End of Sample Question Set -----